CIRC HQ STUDENT HANDOUT | GRADE 6

Tank 7: Plan a fair test

Goal: I can plan a repeatable test that checks what my design claims.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 14, grade 6 path; Playbook B meeting 15, grade 6 path

Ask | Plan | Build | Test | Improve

You are done when

Explain your test, what stays the same, and what result would count as success.

You will use

- Prototype
- Paper
- Pencils
- Ruler or timer if needed
- Reusable test objects

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Compare two tests that change several things.
- Watch a fair test keep the object, starting point, and action consistent.

4 Step 4: Make it work — 17 min

- Write or draw what the product claims to do.
- Choose one thing to count, measure, or observe.
- Plan three trials with the same starting conditions.

5 Step 5: Check one thing — 5 min

- Walk through the test without rushing.
- Ask a partner what must stay the same.
- Get the teacher's approval before the real trials.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Explain your test, what stays the same, and what result would count as success.

What to turn in

A written test plan: what changes and what stays the same.

No computer? Another way to do it

Explain the test aloud to the teacher.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Only conduct teacher-approved tests at table level. No people, valuables, or live animals are test objects.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.